

Anit Paul

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Aspiring AI and Data Science professional with a solid foundation in machine learning, data analysis and model development.
Driven to solve real-world problems using intelligent, data-driven solutions.



Education

BRAC University <i>Bachelor of Science in Computer Science (CGPA: 3.76 / 4.00)</i>	2025 Dhaka, Bangladesh
• Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Image Processing, Computer Vision, Computer Networks, Artificial Intelligence, Computer Security, Ethical Hacking, Software Engineering	
Cumilla Govt. College <i>Science (GPA: 5.00)</i>	2020 Cumilla, Bangladesh
Cumilla Zilla School <i>Science (GPA: 5.00)</i>	2018 Cumilla, Bangladesh

Projects

Real-Time Target Detection and UAV Control for Enhanced Battlefield Situational Awareness

- Developed a real-time UAV surveillance system using deep learning models (YOLOv8, ResNet, VGG) for precise object detection and classification in complex environments.
- Integrated RGB, LiDAR, and infrared sensor data with optimized training pipelines leveraging transfer learning, data augmentation, and TensorFlow frameworks.
- Deployed lightweight, GPU-accelerated models on UAV hardware to enable low-latency inference and robust field performance.

EshopBD

- Built a full-stack e-commerce platform using Laravel (backend), React (frontend), and MySQL for seamless user experience and efficient product management.
- Implemented secure user authentication, product CRUD operations, dynamic search/filtering, and responsive UI to support modern shopping features.
- Designed RESTful APIs for scalable client-server communication and optimized database queries to ensure fast load times and smooth performance.

Heart Disease Classification

- Preprocessed clinical data by handling missing values, encoding categorical variables, and applying feature scaling to prepare for classification tasks.
- Implemented and compared Logistic Regression and XGBoost models to predict heart disease presence using performance metrics such as accuracy, precision, recall, and F1-score.
- Visualized model results with confusion matrices, ROC curves, and AUC scores using Matplotlib and Seaborn for effective performance interpretation.

Mushroom Species Type

- Processed and analyzed mushroom dataset to classify species as poisonous or edible, using both supervised (Logistic Regression, XGBoost, Decision Tree) and unsupervised (KMeans) learning algorithms and then tried with Neural Network.
- Engineered features and applied label encoding, normalization, and exploratory data analysis to improve model interpretability and performance.
- Evaluated models using accuracy, precision, recall, and F1-score to compare performance across classifiers and fine-tuned hyperparameters for optimal results.

Technical Skills

Languages: Python, Go, Javascript, TypeScript, C

Technologies: React.js, Angular, Vue.js, Django, Flask, Express.js, TensorFlow, PyTorch, Bootstrap, Laravel

Concepts: Compiler Design, Operating System, Computer Network, Encryption, Decryption, Artificial Intelligence, Machine Learning, Neural Networks, Computer Vision, API, Database Normalization, Agile Methodology, Cloud Computing

Certificates

Highlighted Certifications:

- **AI Engineer for Data Scientists Associate** – *DataCamp, July 2025*

Other Certifications:

- **Fundamentals of Artificial Intelligence & Cyber Security**
- **Mastering Network Security: Defending Against Cyber Threats**
- **Ethical Hacking: Linux Privilege Escalation**
- **CyberChef Basics for Beginners**
- **Understanding Artificial Intelligence**
- **Introduction to Python**
- **Intermediate Python**
- **Data Manipulation with pandas**
- **Joining Data with pandas**
- **Statistics in Python**
- **Data Visualization with Matplotlib**
- **Data Visualization with Seaborn**

Papers & Publications

Countering RF Jamming in FPV Drones: A Hybrid Strategy for Target Tracking and Pathway Optimization
(Ongoing Thesis)

- Proposed a unified framework combining RF jamming detection, computer vision-based tracking and adaptive path planning to ensure FPV drone resilience under electronic warfare conditions.

Advancement in Detection of Melanoma: A Systematic Literature Review on CNN Utilization
(Ongoing)

- Reviewed recent CNN-based approaches for melanoma detection, highlighting architecture choices, dataset usage, and key research challenges in clinical application.

Awards

6th Place, Hack the AI Contest Hackathon AI Network Control Agent with Smythos
Year: 2025

- Built an AI-powered terminal agent using Smythos and LLMs to observe, block, unblock, and manage websites within a network using natural language commands.
- Integrated secure and intuitive user interaction to control network operations with plain-text prompts, demonstrating practical AI-agent deployment.

Champion, National High School Science Fair Blind Navigation System
Year: 2018

- Developed Arduino-based portable sunglasses for blind people featuring robust object detection using sonar technology.

18th Place, National High School Programming Quiz Contest 2017

5th Place, National High School Programming Quiz Contest 2018